PHIVOLCS Special Report – The 27 July 2019 Magnitudes 5.4, 5.9, 5.8 Itbayat, Batanes Earthquake

Epicenter/s: At 20.98°N, 122.21°E (5.4 Itbayat, Batanes Earthquake)

Note: The reported intensities of each of 3 three events are all the same (and so the resulting calculations are the same), and they also have close epicenters. We will only consider the first epicenter.

27 July 2019 04:16 AM	20.98	122.21	43 km, N 62°E of Itbayat, Batanes	5.4	15	***Earthquake Information 4:*** ***Intensity VI*** – Itbayat, Batanes ***Intensity IV*** – Basco, Ivana, and Mahatao, Batanes ***Intensity III*** – Uyugan and Sabtang, Batanes Instrumental Intensity: ***Intensity III*** – Basco, Batanes	***Intensity VI*** – Brgys. Santa Maria, Santa Lucia, Santa Rosa, and San Rafael, Itbayat, Batanes ***Intensity V*** – Brgy. Raele, Itbayat, Batanes ***Intensity IV*** – Basco, Ivana, and Mahatao, Batanes ***Intensity III*** – Uyugan and Sabtang, Batanes Instrumental Intensity: ***Intensity III*** – Basco, Batanes

Intensity VI

Places	Coordinates	Findings (from the Method)
Santa Maria, Batanes	20.7731° N, 121.8620° E	
Santa Lucia, Batanes	20.7556° N, 121.8406° E	
Santa Rosa, Batanes	20.8260° N, 121.8620° E	
San Rafael, Batanes	20.7589° N, 121.8278° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 20.792707° N, 121.834160° E
Epicenter 1σ interval: (H ± 0.008579°, K ± 0.008108°)
Distance to PHIVOLCS Measured Epicenter: 44.252 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.034052	3.624
P2	0.037662	4.180
P3	0.043399	4.699
P4	0.034400	3.817

Minimum R: 0.034052° (~3.624 km)
Maximum R: 0.043399° (~4.699 km)
Mean R_prox: 0.037378°
Radius std: 0.003751°

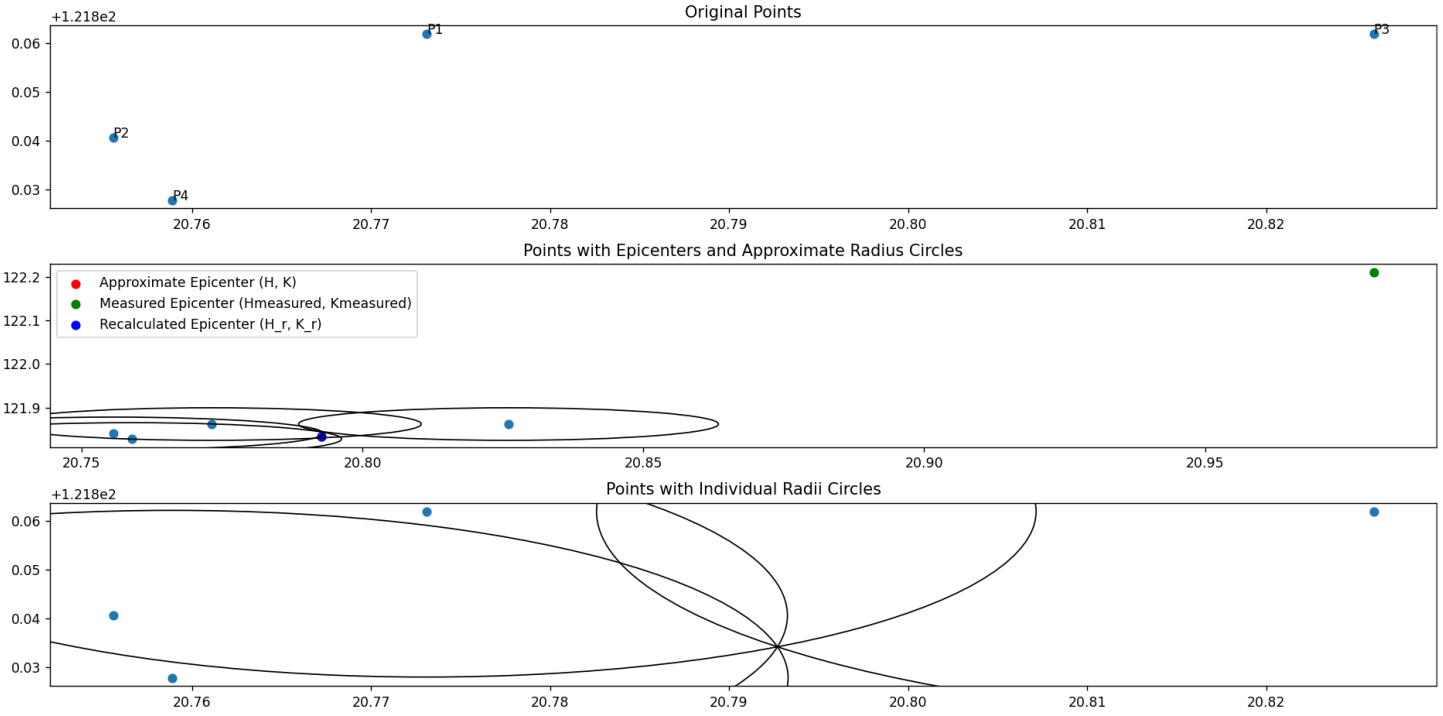
2) Recalculated Epicenter (After Outlier Removal)

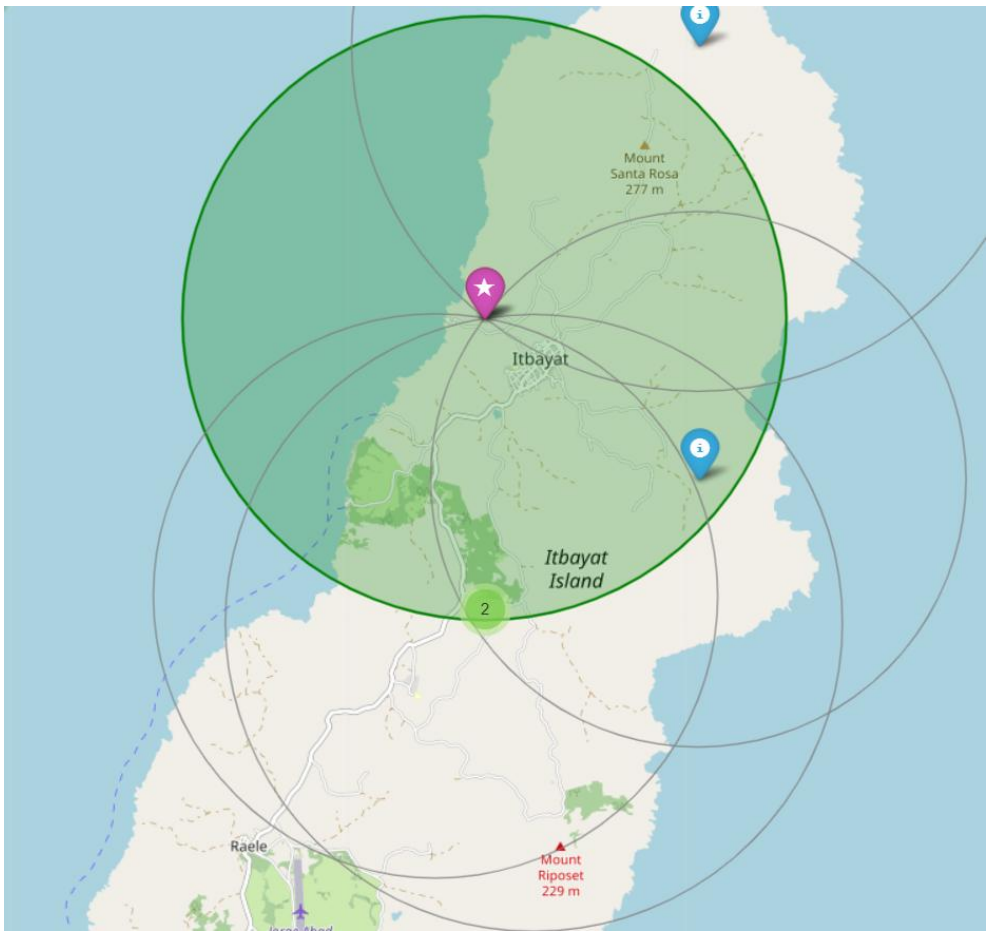
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Minimum R: 0.034052° (~3.624 km)
Maximum R: 0.043399° (~4.699 km)


Original Points
+1.218e2
P1
P2
P3
P4
0.06
0.05
0.04
0.03
20.76
20.77
20.78
20.79
20.80
20.81
20.82
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
122.2
122.1
122.0
121.9
20.75
20.80
20.85
20.90
20.95
Points with Individual Radii Circles
+1.218e2
0.06
0.05
0.04
0.03
20.76
20.77
20.78
20.79
20.80
20.81
20.82


Mount
Santa Rosa
277 m
Itbayat
Itbayat
Island
2
Raele
Mount
Riposet
229 m

Intensity IV

Places	Coordinates	Findings (from the Method)
Basco, Batanes	20.4634° N, 122.0042° E	
Ivana, Batanes	20.3781° N, 121.9302° E	
Mahatao, Batanes	20.4052° N, 121.9530° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 18.743910° N, 123.900099° E

Epicenter 1σ interval: (H ± 0.000000°, K ± 0.000000°)

Distance to PHIVOLCS Measured Epicenter: 305.055 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	2.559508	275.667
P2	2.559508	274.985
P3	2.559508	275.198

Minimum R: 2.559508° (~275.667 km)

Maximum R: 2.559508° (~274.985 km)

Mean R_prox: 2.559508°

Radius std: 0.000000°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 18.743910° N, 123.900099° E

Epicenter 1σ interval: (H ± 0.000000°, K ± 0.000000°)

Distance to PHIVOLCS Measured Epicenter: 305.055 km

Radii to (H, K)

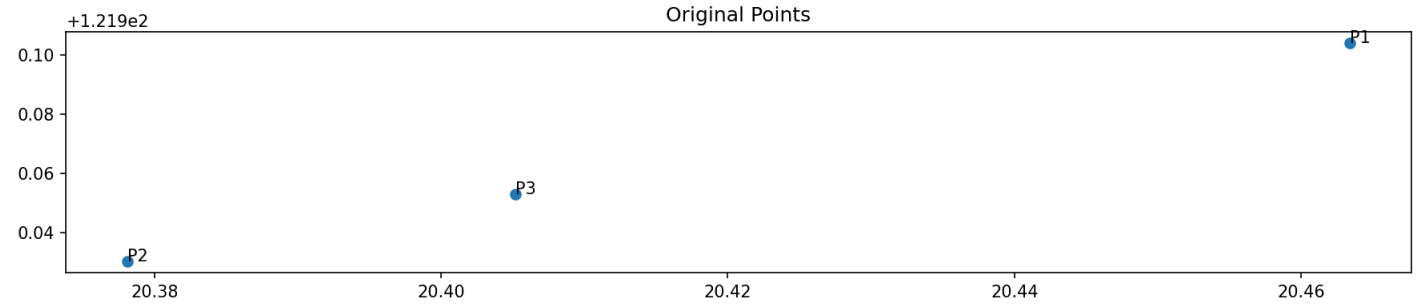
Point	Radius (deg)	Radius (km)
P1	2.559508	275.667
P2	2.559508	274.985
P3	2.559508	275.198

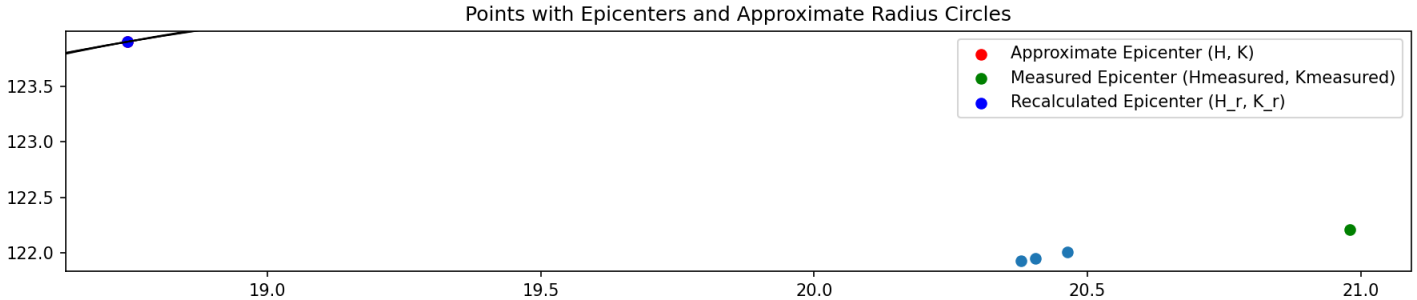
Minimum R: 2.559508° (~275.667 km)

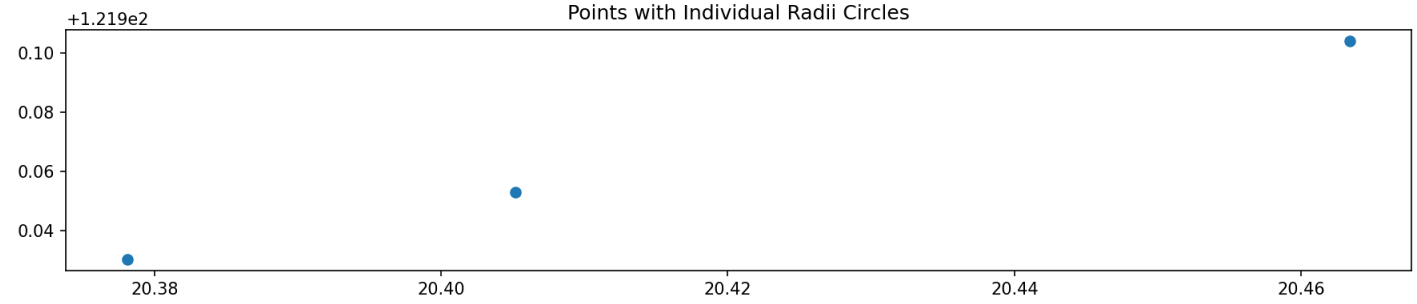
Maximum R: 2.559508° (~274.985 km)

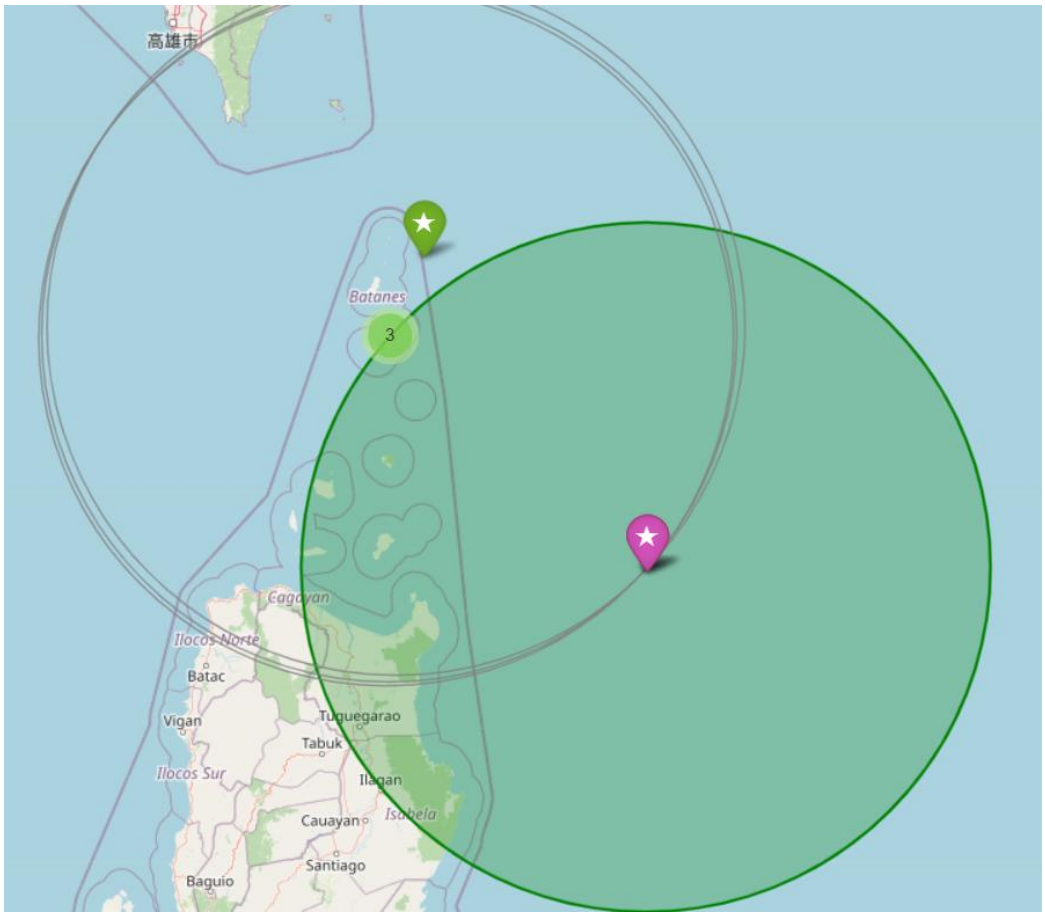
Mean R_prox: 2.559508°

Radius std: 0.000000°


Original Points
+1.219e2
0.10
0.08
0.06
0.04
P1
P2
P3
20.38
20.40
20.42
20.44
20.46


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
123.5
123.0
122.5
122.0
19.0
19.5
20.0
20.5
21.0


Points with Individual Radii Circles
+1.219e2
0.10
0.08
0.06
0.04
20.38
20.40
20.42
20.44
20.46


高雄市
Batanes
3
Cagayan
Ilocos Norte
Batac
Vigan
Tuguegarao
Tabuk
Ilocos Sur
Ilagan
Isabela
Cauayan
Santiago
Baguio

Epicenter/s: At 21.07°N, 121.96°E (5.9 Itbayat, Batanes Earthquake)

27 July 2019 07:37 AM	21.07	121.96	33 km, N 22°E of Itbayat, Batanes	5.9	8	***Earthquake Information 4:*** *Intensity VII* – Itbayat, Batanes *Intensity V* – Basco, Ivana, and Mahatao, Batanes *Intensity IV* – Uyugan and Sabtang, Batanes Instrumental Intensity: *Intensity IV* – Basco, Batanes	***Intensity VII*** – Brgys. Santa Maria, Santa Lucia, Santa Rosa, and San Rafael, Itbayat, Batanes ***Intensity VI*** – Brgy. Raele, Itbayat, Batanes ***Intensity V*** – Basco, Ivana, and Mahatao, Batanes ***Intensity IV*** – Uyugan and Sabtang, Batanes Instrumental Intensity: ***Intensity IV*** – Basco, Batanes

Intensity VII

Places	Coordinates	Findings (from the Method)
Santa Maria, Batanes	20.7731° N, 121.8620° E	
Santa Lucia, Batanes	20.7556° N, 121.8406° E	
Santa Rosa, Batanes	20.8260° N, 121.8620° E	
San Rafael, Batanes	20.7589° N, 121.8278° E	

Intensity V

Places	Coordinates	Findings (from the Method)
Basco, Batanes	20.4634° N, 122.0042° E	
Ivana, Batanes	20.3781° N, 121.9302° E	
Mahatao, Batanes	20.4052° N, 121.9530° E	

Epicenter/s: At 20.89°N, 121.84°E (5.8 Itbayat, Batanes Earthquake)

27 July 2019 09:24 AM	20.89	121.84	11 km N 02°W of Itbayat, Batanes	5.8	11	***Earthquake Information 3:*** *Intensity IV* – Itbayat, Basco, Ivana and Mahatao, Batanes *Intensity III* – Sabtang and Uyugan, Batanes Instrumental Intensity: *Intensity IV* – Basco, Batanes	*Intensity VII* – Brgys. Santa Maria, Santa Lucia, Santa Rosa and San Rafael, Itbayat, Batanes *Intensity VI* – Brgy. Raele, Itbayat, Batanes *Intensity V* – Basco, Ivana and Mahatao, Batanes *Intensity IV* – Uyugan and Sabtang, Batanes Instrumental Intensity: *Intensity IV* – Basco, Batanes

Intensity VII

Places	Coordinates	Findings (from the Method)
Santa Maria, Batanes	20.7731° N, 121.8620° E	
Santa Lucia, Batanes	20.7556° N, 121.8406° E	
Santa Rosa, Batanes	20.8260° N, 121.8620° E	
San Rafael, Batanes	20.7589° N, 121.8278° E	

Intensity V

Places	Coordinates	Findings (from the Method)
Basco, Batanes	20.4634° N, 122.0042° E	
Ivana, Batanes	20.3781° N, 121.9302° E	
Mahatao, Batanes	20.4052° N, 121.9530° E	

CONSOLIDATED TABLE				
Intensity	Initial Epicenter	Recalculated Epicenter	Outliers	Distance Error
Actual Location	20.98 N, 122.21° E	-	-	-
VI	20.7927° N, 121.8342° E	20.7927° N, 121.8342° E	None	44.252 km
IV	18.7439° N, 123.9001° E	18.7439° N, 123.9001° E	None	305.055 km

INTENSITY-BY-INTENSITY INTERPRETATION (BATANES EARTHQUAKE)

INTENSITY VII / VI — Itbayat (Very Destructive / Destructive Shaking)

Affected localities:

- Santa Maria
- Santa Lucia
- Santa Rosa
- San Rafael

Interpretation:
All these barangays are located on the **same small island (Itbayat)** and only **1–3 km apart**. This creates a **high-intensity cluster with almost no spatial spread**.

Physically, this means:

- These communities all sit very close to the fault rupture.
- Their intensities reflect **direct near-field damage**: collapsed stone houses, church damage, and landslides.
- However, because they lie in a compact group, their reported intensities **cannot triangulate the source direction**.

In macroseismic inversion, this produces:

- **Excellent information on severity**,
- **Inferior information on direction**,
- Resulting in epicenter estimates that drift anywhere within a **broad zone around Itbayat**, often 30–60 km from the actual source.

Physical conclusion:
Intensity VI/VII confirms that the quake was very near Itbayat. Still, because all strong-motion reports come from nearly the exact coordinates, they lack geometric leverage to pinpoint the precise epicenter offshore.

INTENSITY V — Batan Island (Strong Shaking)

Affected localities:

- Basco
- Ivana
- Mahatao

Interpretation:
These towns lie on **Batanes Island**, ~80–100 km south of Itbayat.
Intensity V here represents:

- moderate structural vibrations,
- swaying of buildings,
- minor cracks,
- alarmed residents.

The key physical insight is that these towns are:

- **much further from the rupture**,
- positioned along a **southward line**,
- nearly **collinear** on the island's western coast.

This produces two physical effects:

1. Intensity V no longer reflects precise source direction.

By this distance, intensity is dominated by:

- wave attenuation,
- local soil conditions,

- basin effects,
- building types.

2. Geometric instability arises from collinearity.

Because Basco–Mahatao–Ivana lie almost in a straight line, any attempt to reconstruct an epicenter from these intensities produces:

- large circle radii,
- unstable intersections,
- epicenters that may fall **hundreds of kilometers away**.

Physical conclusion:
Intensity V confirms regional shaking reached Batan Island, but because the points are far from the source and aligned, they physically **cannot localize the epicenter**.

INTENSITY IV / III — Regional Weak to Moderate Shaking

Depending on the report, some areas of Batan Island and nearby provinces felt Intensity IV–III.

Interpretation:
These intensities represent shaking that is:

- weak to moderate,
- heavily influenced by building type, soil type, and subjective perception,
- easily distorted by local amplification.

At this level, **distance from the source is not the dominant factor**.

Thus, any inversion using these far-field intensities produces:

- enormous radii (150–300 km),
- wide uncertainty,
- no meaningful epicentral constraint.

Physical conclusion:
Low intensities in Batan or nearby islands only confirm that the earthquake was perceptible regionally; they **provide no directional information** about the source.

CONSOLIDATED PHYSICAL INTERPRETATION (BATANES)

The Batanes earthquake exhibits a **physically predictable but inversion-challenging** macroseismic pattern because of the **unique geography of the islands**.

1. Strong-motion reports are tightly clustered

All Intensity VI/VII stations are on a **single, tiny island** (Itbayat).
This cluster provides:

- **accurate constraint on severity**,
- **poor constraint on location** (no spread = no triangulation).

2. Moderate-motion reports come from another island

Intensity V is observed on Batan Island:

- far from the source,
- nearly collinear,
- strongly influenced by local soil effects.

Thus, these do not help localize the epicenter and may push solutions southward or outward.

3. Low intensities (IV, III) only define the perceptibility boundary

These intensities contribute to understanding the **extent** of shaking, but not the **source**.

4. The epicenter lies offshore north of Itbayat

Because:

- Itbayat has all the high intensities, but
- no distribution to the north, east, or west,

Any circle-based inversion will locate the epicenter somewhere around the **centroid of the high-intensity cluster**, but not precisely where the seismological solution lies (offshore).

5. This event demonstrates a fundamental physical limitation

Macroseismic epicenter estimation requires:

- **multiple** high-intensity points
- **separated** in space
- surrounding the epicenter in **more than one direction**

Batanes, being a small archipelago, **cannot provide this geometry**.

Therefore:

- High intensities confirm proximity to Itbayat
- Low intensities outline regional reach
- But the exact strike point offshore cannot be inferred purely from intensity geometry

This is a **geometric limitation**, not a methodological weakness.

Summary:

The Batanes earthquake's macroseismic field correctly indicates strong shaking centered on Itbayat, but because all severe intensities are concentrated on a single small island and moderate intensities are located far away on a collinear island chain, the physical geometry prevents precise epicenter localization from intensities alone.